MATE

New combination creates new value

新しい組み合わせが、新しい価値を生みだす。

機能性複合材料

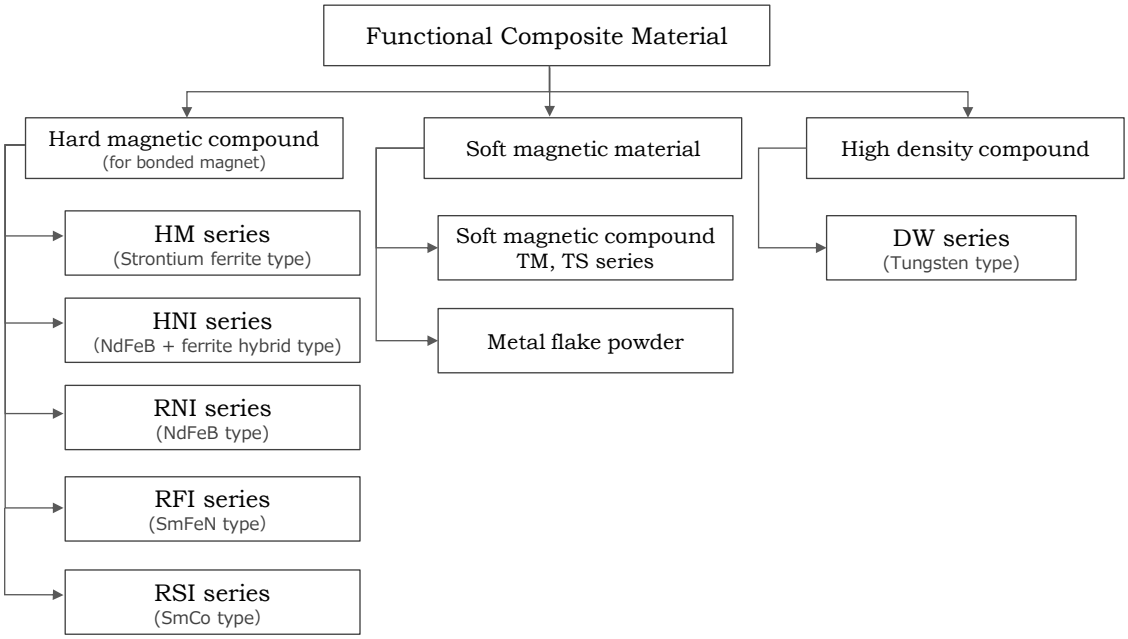
Functional Composite Material

What is Functional Composite Material?

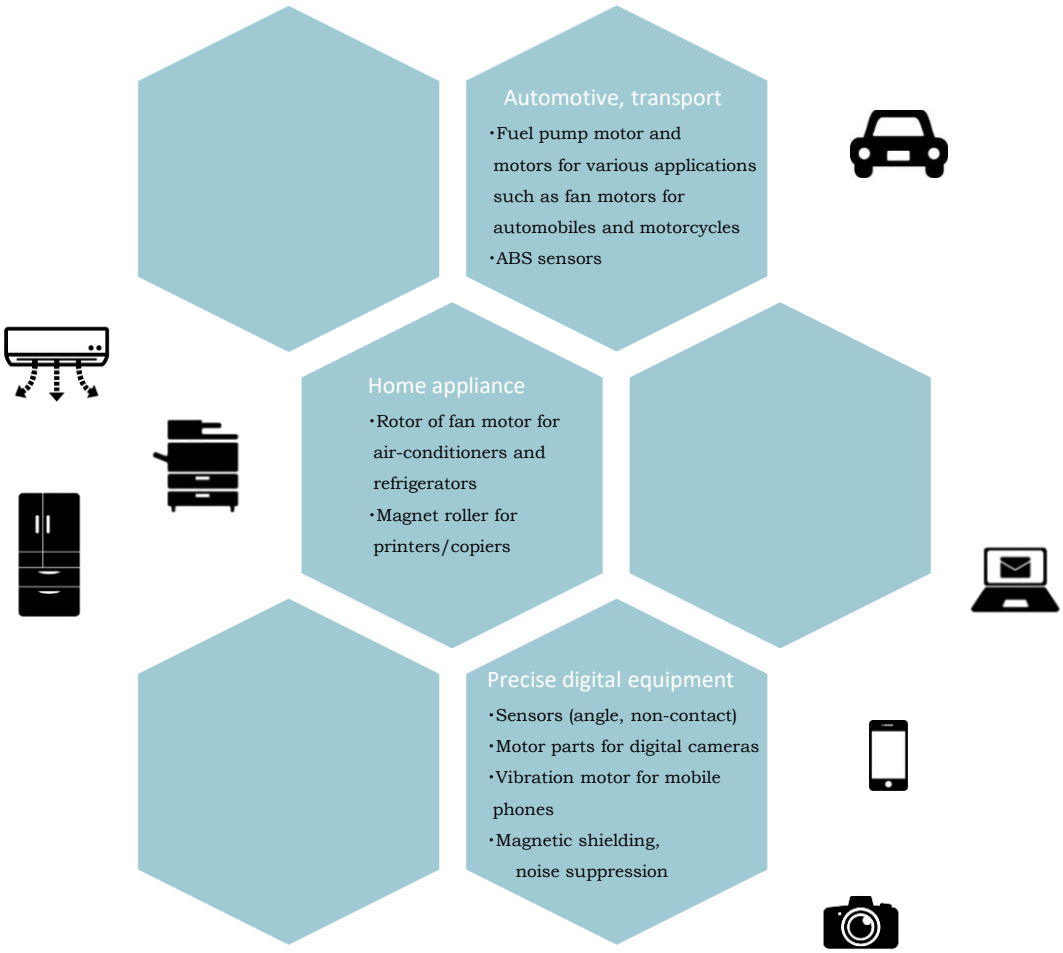
Functional Composite Materials are made by compounding a resin and filler for injection and extrusion molding. We can control magnetic properties, physical properties and moldability utilizing our own technologies in surface treatment, recipe and kneading. We have various filler and resin options allowing us to choose the best to satisfy each application (optimize heat resistance, chemical stability, strength, flexibility and dimensional stability). Such compounds have excellent moldability. Therefore small, complex and thin shape of magnets can be realized (insert, outsert, or integrated molding is possible and assembly process can be rationalized).



Line-up



Applications



Our functional composite materials are widely used in other various applications utilizing their functions and properties.

Compounds for bonded magnet are the materials made by hard magnetic powder and thermoplastic resin suitable for injection molding.

1. Feature

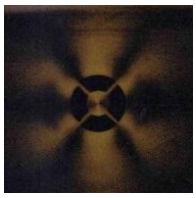
We control magnetic, electrical and physical properties, and moldability through our proprietary technologies in surface treatment, formulation and kneading.
We prepare various kinds of magnetic powders and resins and can choose the best ones for each application (to optimize heat resistance, chemical stability, strength, flexibility and dimensional stability).
Compounds can achieve compact, complex or thin shapes because of their excellent moldability (insert, outsert and integrated molding can be done, and rationalization in assembly process can be done).

2. Applications

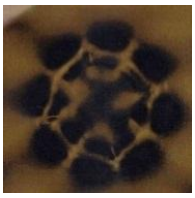
Motors and sensors for home appliances and OA equipment, Magnet rollers for printers and copiers, motors and sensors for automotive, various compact motors.

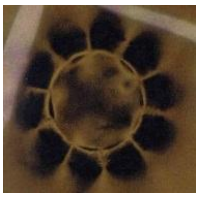
3. Examples of magnetization

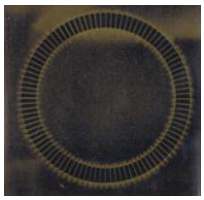
Possible to design magnetic wave form freely (multi-pole, radial magnetization, and make a magnet with required magnetic force, orientation and magnetic pattern).

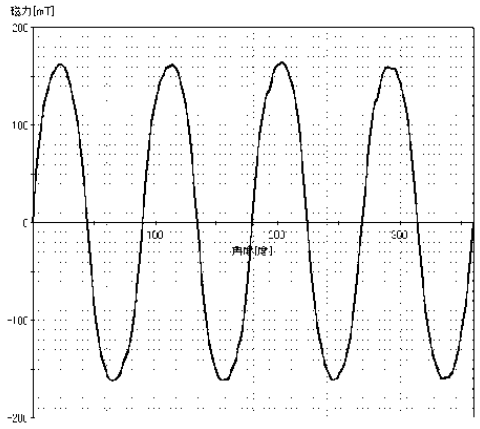

Surface 4 poles

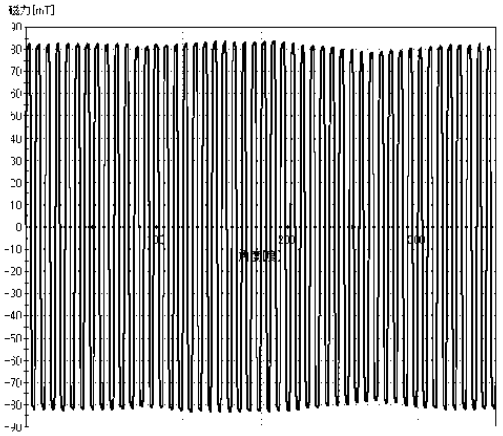

Inner 6 poles


Outer 8 poles


Outer 10 poles


Surface 96 poles





4. Product Line-up

HM series

(Sr ferrite base)
The best material in terms of cost/performance. Chemically stable because of oxide.

HNI series

(NdFeB & Sr ferrite base hybrid)
Hybrid materials with rare earth NdFeB and Sr ferrite are dispersed homogeneously and with features of both NdFeB and Sr ferrite.

RNI series

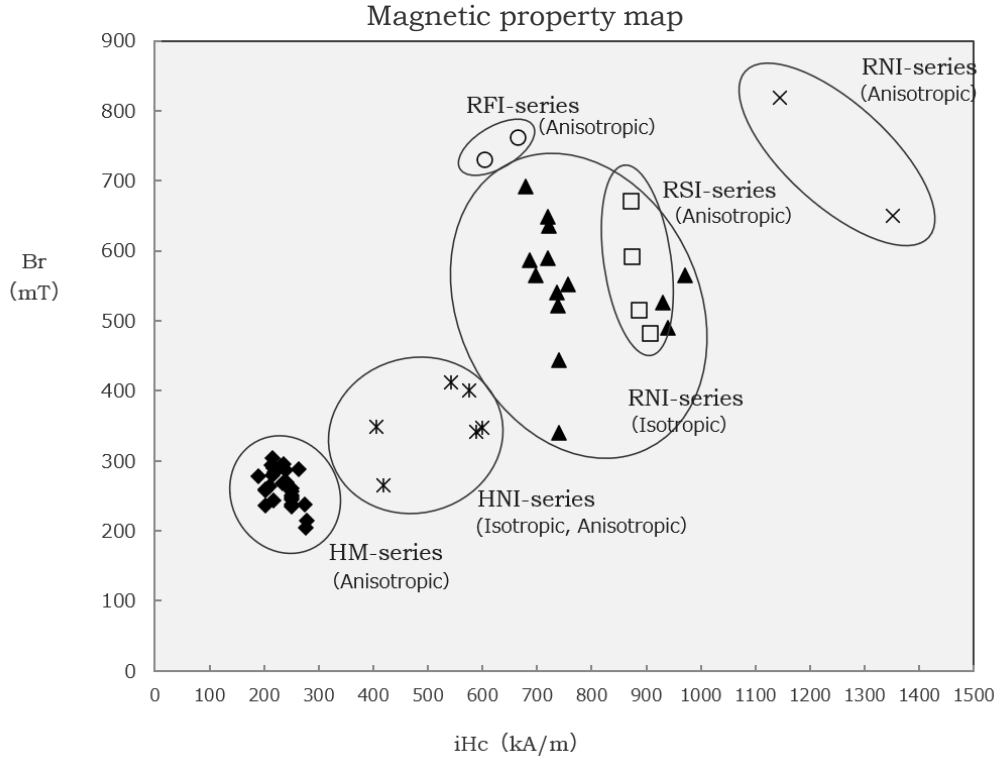
(Rare earth NdFeB base)
The highest energy product (BHmax) among materials for bonded magnets and widely used in leading-edge products.
Makes magnets small and light with several-fold magnetic force compared to ferrite base material.

RFI series

(Rare earth SmFeN base)
It possesses magnetic strength approaching that of neodymium anisotropic materials and offers superior corrosion resistance compared to neodymium magnets.

RSI series

(Rare earth SmCo base)
Better anti-oxidation nature compared with NdFeB based bonded magnets. Also excellent heat resistance. These have small thermal demagnetization and are suitable for severe operating environments like automotive and for areas requiring high reliability.



5. Properties (values are typical and not guaranteed)

The listed grades are just typical ones from our product range.
We can adjust properties according to specific requirements from customers.
Data not listed in the following tables such as water/heat resistance and tensile strength are available on request.
Please refer to our inquiry page at the end of the catalogue or directly to our web site.

HM series (Sr ferrite base)

			JIS C2501				ASTM D-792	ASTM D-1238	ASTM D-790	ASTM D-256	JIS K7197
		Unit	mT	kA/m	kA/m	kJ/m^3	g/cm^3	g/10min	MPa	kJ/m^2	×10^{-5}/℃
	Binder	Grade	Br	bHc	iHc	BHmax	Molded Density	MFR Melt Flow Rate	Flexural Strength	IZOD Impact Strength (Unnotched)	Coefficient of linear thermal expansion
Anisotropic	PA6	HM-1114SD	238	182	274	11.2	3.25	134*	198	37.8	–
		HM-1115	251	190	251	12.4	3.36	115*	191	30.2	–
		HM-1116	260	195	250	13.3	3.45	90*	195	27.0	–
		HM-1117	270	195	242	14.2	3.53	93*	193	24.0	–
		HM-1118H	280	181	215	15.3	3.60	120*	174	18.3	3.1 (-40～120℃)
		HM-1119H	284	189	219	15.8	3.66	125*	171	15.0	–
		HM-1120H	286	198	240	16.1	3.65	147*	180	19.0	2.5 (-40～120℃)
		HM-1121	294	193	214	17.0	3.76	80*	170	14.1	–
		HM-1122B	296	200	235	17.2	3.74	155*	170	15.0	–
Anisotropic	PA12	HM-1209	204	161	277	8.3	2.90	35*[5]	120	–	–
		HM-1210K	214	165	278	9.0	3.00	177*	128	50.0	6.5 (-40～110℃)
		HM-1213	235	177	251	10.9	3.21	130*	125	28.0	5.2 (-40～120℃)
		HM-1213X	238	177	250	11.1	3.23	200*	133	31.0	–
		HM-1215	245	185	251	11.9	3.30	110*	129	26.5	–
		HM-1216	257	192	251	13.0	3.41	90*	128	21.4	–
		HM-1218	268	191	234	14.1	3.50	78*	129	18.0	4.2 (-40～120℃)
		HM-1220KD8	288	209	263	16.4	3.64	160*	101	17.7	–
		HM-1220S	290	193	218	16.6	3.69	77*	133	15.3	–
		HM-1220X	293	200	230	16.8	3.72	100*	120	14.5	3.6 (-40～110℃)
		HM-1222N	304	192	216	18.0	3.81	137*	93	14.4	–
Anisotropic	PPS	HM-1613	236	171	203	11.0	3.35	114*[1]	120	11.9	–
		HM-1614X2	243	175	217	11.6	3.40	142*[4]	134	12.2	–
		HM-1615X2	248	180	250	12.0	3.47	106*[4]	137	11.5	–
		HM-1616	258	169	202	13.1	3.55	115*[1]	94	8.1	2.1 (-40～150℃)
		HM-1616X	259	174	202	13.1	3.54	200*[1]	95	8.0	–
		HM-1617H	263	180	210	13.5	3.59	125*[1]	100	7.9	–
		HM-1618X	266	187	240	13.8	3.62	140*[1]	121	9.0	–
		HM-1619	278	165	190	15.0	3.72	125*[1]	95	7.4	1.9 (-40～120℃)

HNI series (NdFeB and Sr ferrite base hybrid)

			BMG-2001 20190111				ASTM D-792	ASTM D-1238	ASTM D-790	ASTM D-256	JIS K7197
		Unit	mT	kA/m	kA/m	kJ/m^3	g/cm^3	g/10min	MPa	kJ/m^2	×10^{-5}/℃
	Binder	Grade	Br	bHc	iHc	BHmax	Molded Density	MFR Melt Flow Rate	Flexural Strength	IZOD Impact Strength (Unnotched)	Coefficient of linear thermal expansion
Isotropic	PA12	HNI-B235L1	341	222	589	20.1	4.39	340*[2]	114	9.9	–
Anisotropic			412	239	542	29.5					
Isotropic	PPS	HNI-B625	265	175	419	12.1	4.07	380*[1]	138	8.7	–
Anisotropic			349	224	405	22.5					
Isotropic		HNI-B635	347	231	599	20.9	4.37	430*[1]	137	8.2	–
Anisotropic			401	246	576	28.6					–

RNI series (Rare earth NdFeB base)

			BMG-2001 20190111				ASTM D-792	ASTM D-1238	ASTM D-790	JIS K7191-1,2	JIS K7197
	Unit		mT	kA/m	kA/m	kJ/m^3	g/cm^3	g/10min	MPa	℃	×10^{-5}/℃
	Binder	Grade	Br	bHc	iHc	BHmax	Molded Density	MFR Melt Flow Rate	Flexural Strength	Deflection temp. under load (1.8MPa)	Coefficient of linear thermal expansion
Isotropic	PA12	RNI-25LF2	340	243	740	20.8	4.03	240*2	125	-	2.8 (-40～110℃)
		RNI-40LF	444	307	740	34.3	4.51	330*2	123	-	-
		RNI-55LF	522	352	738	46.2	5.07	540*2	81	-	-
		RNI-60HR	541	361	737	49.7	5.04	400*3	99	133	-
		RNI-60LF	552	370	756	51.5	5.28	420*2	107	-	-
		RNI-70	589	383	720	58.0	5.35	340*2	95	-	-
		RNI-80HR	636	410	722	67.0	5.59	250*3	72	-	-
		RNI-90HR	648	416	720	69.0	5.66	200*3	73	142	-
		RNI-95HR	692	428	680	76.6	5.98	130*3	67	-	1.9 (-40～120℃)
Isotropic	PPS	RNI-4650	490	345	940	42.2	4.94	800*1	107	-	-
		RNI-4660	526	365	930	48.0	5.20	460*1	104	-	-
		RNI-PS65H	565	377	697	53.9	5.20	230*4	89	-	1.6 (-50～100℃)
		RNI-4670	565	394	972	55.5	5.44	500*1	82	-	-
		RNI-PS70H	587	389	686	57.8	5.33	320*1	83	241	1.4 (-40～150℃)
Anisotropic	PA12	RNI-5214A3	812	545	1170	118.4	5.14	360*2	101	-	-
Anisotropic	PPS	RNI-5610V	663	470	1370	81.1	4.65	500*1	85	-	-

RFI series (Rare earth SmFeN base)

			JIS C2501				ASTM D-792	ASTM D-1238	ASTM D-790	ASTM D-256	JIS K7197
	Unit		mT	kA/m	kA/m	kJ/m^3	g/cm^3	g/10min	MPa	kJ/m^2	×10^{-5}/℃
	Binder	Grade	Br	bHc	iHc	BHmax	Molded Density	MFR Melt Flow Rate	Flexural Strength	IZOD Impact Strength (Unnotched)	Coefficient of linear thermal expansion
Anisotropic	PA12	RFI-9211	731	456	603	96.3	4.60	110*	55	-	-
		RFI-9213	763	489	664	104.7	4.68	150*	56	-	-

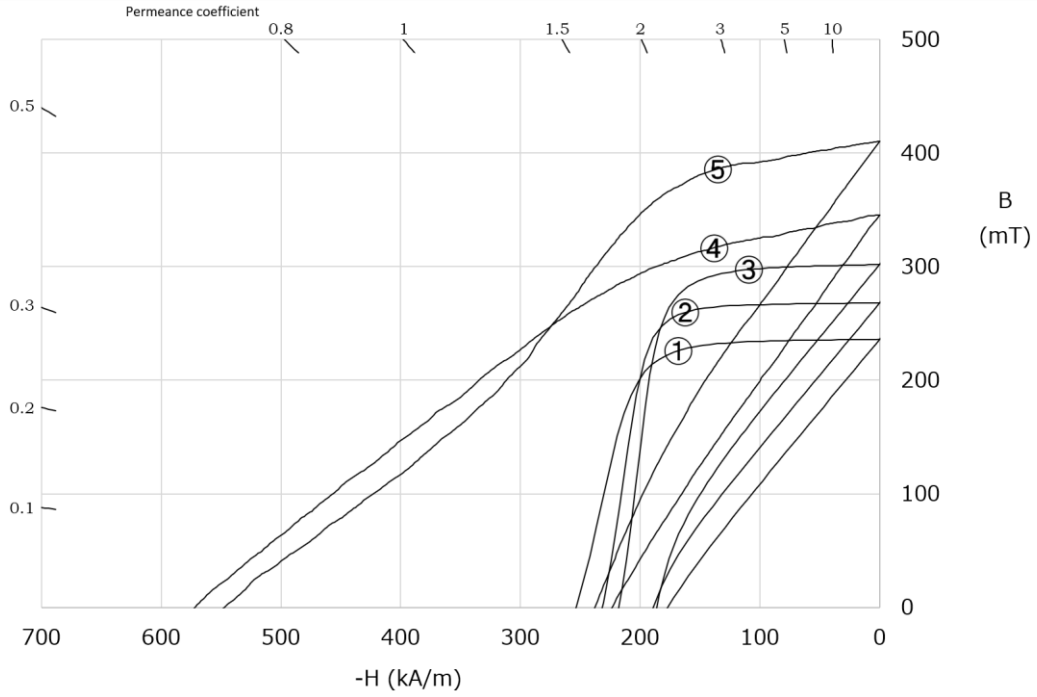
RSI series (Rare earth SmCo base)

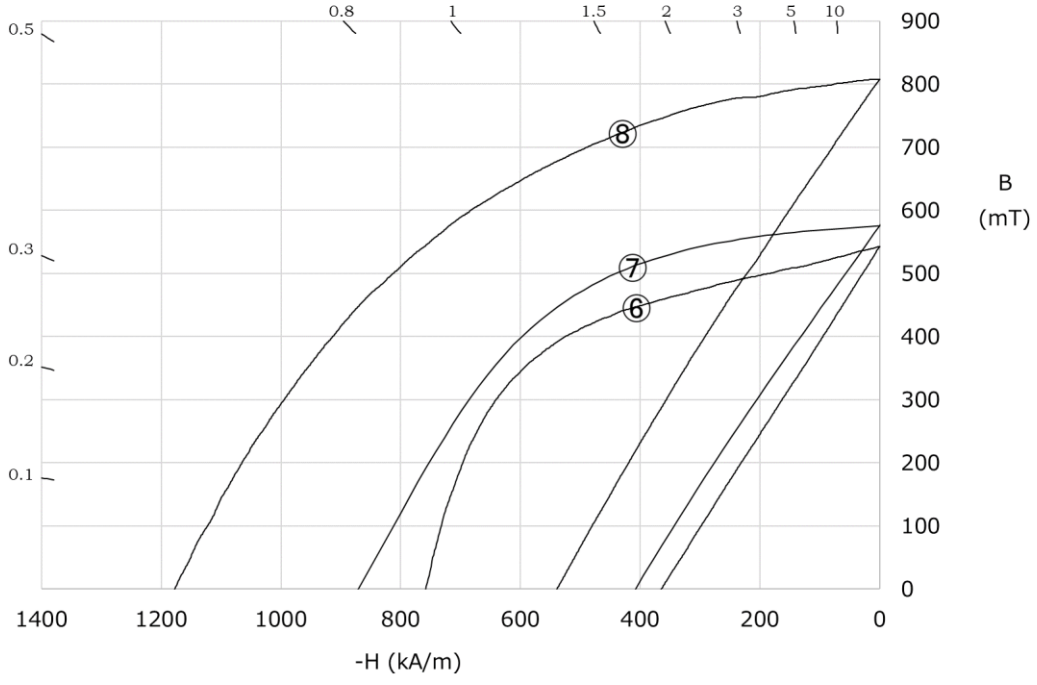
			JIS C2501				ASTM D-792	ASTM D-1238	ASTM D-790	ASTM D-256	JIS K7197
	Unit		mT	kA/m	kA/m	kJ/m^3	g/cm^3	g/10min	MPa	kJ/m^2	×10^{-5}/℃
	Binder	Grade	Br	bHc	iHc	BHmax	Molded Density	MFR Melt Flow Rate	Flexural Strength	IZOD Impact Strength (Unnotched)	Coefficient of linear thermal expansion
Anisotropic	PA12	RSI-8285	592	439	874	68.2	5.26	260*	99	16.6	-
		RSI-8211	673	481	830	86.3	6.19	450*2	78	7.5	-
	PPS	RSI-PS55	483	366	907	45.8	4.63	200*4	78	8.3	-
		RSI-PS60	516	381	886	50.5	5.01	95*4	74	8.0	2.71(23～141℃)

measurement condition of MFR

* 270℃×10kg
*1 330℃×10kg
*2 270℃×5kg
*3 270℃×2.16kg
*4 330℃×5kg
*5 230℃×5kg

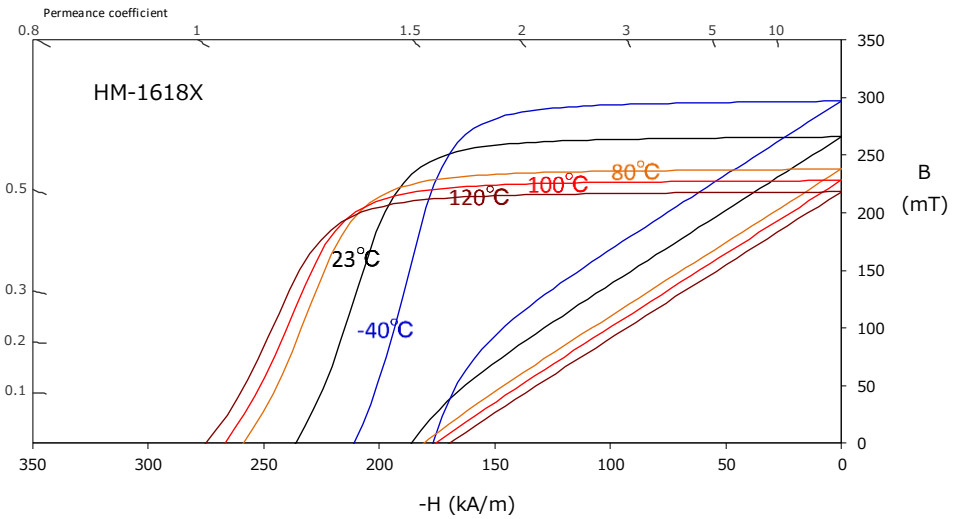
6. Demagnetization curve

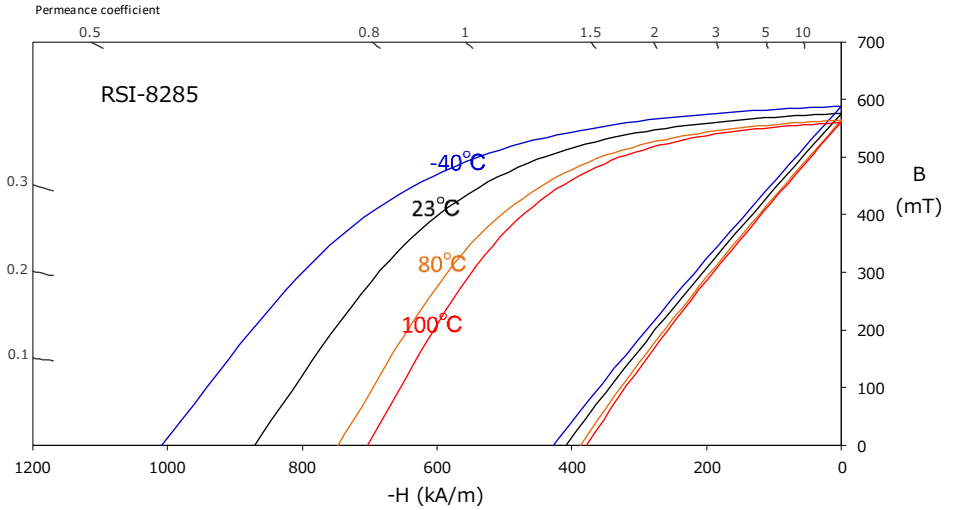


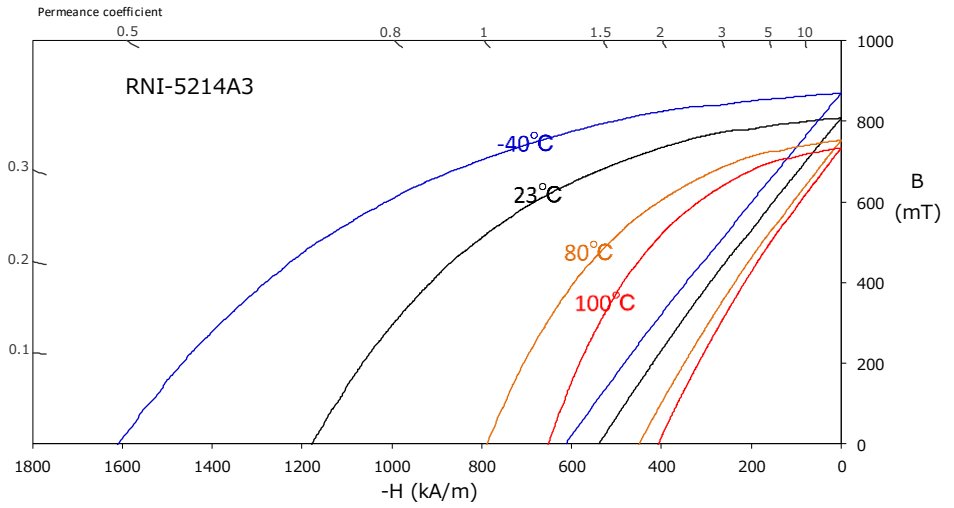


①	HM-1213	⑤	HNI-B235L1（Anisotropic）
②	HM-1218	⑥	RNI-60LF
③	HM-1222N	⑦	RSI-8285
④	HNI-B235L1（Isotropic）	⑧	RNI-5214A3

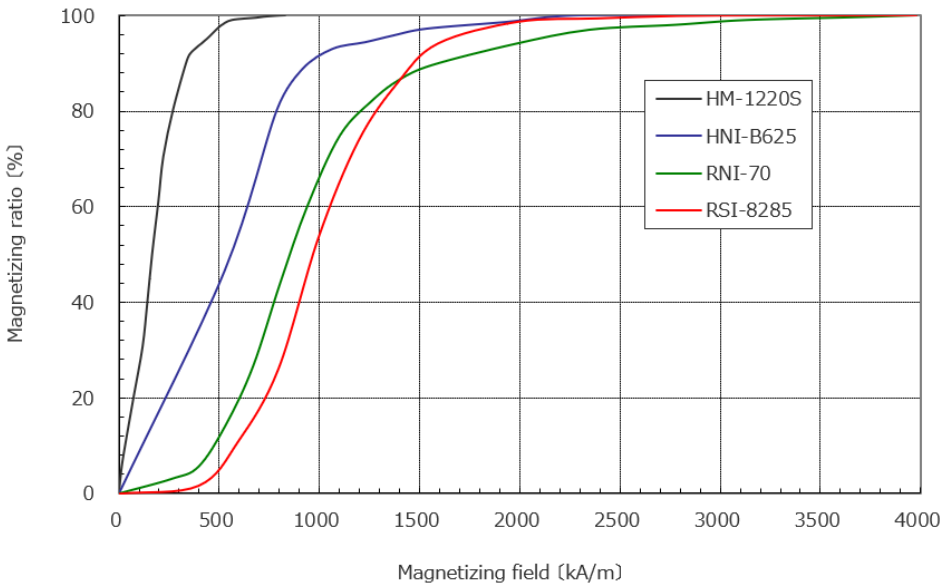
7. Magnetization curve with temperatures



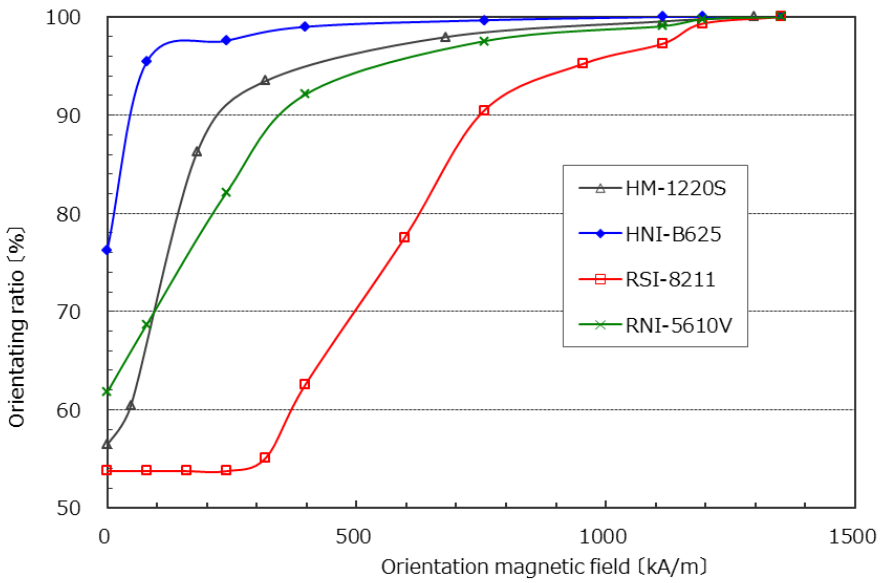




8. Magnetizing properties



9. Orientation properties

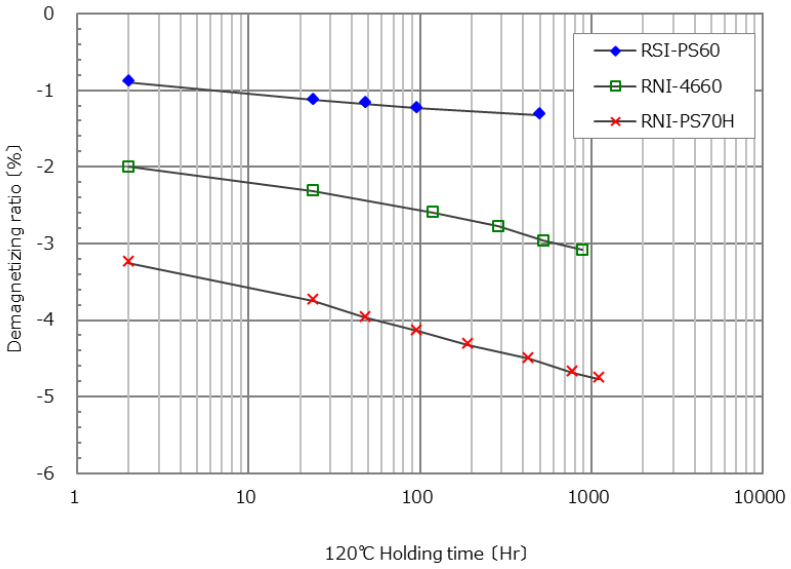


※A value at which flux value is saturated regards as orientating ratio 100%
(Shape : 10×10×7 or φ10×7)

10. Thermal demagnetization properties

Variation against time of irreversible demagnetization ratio (flux) : at 120℃

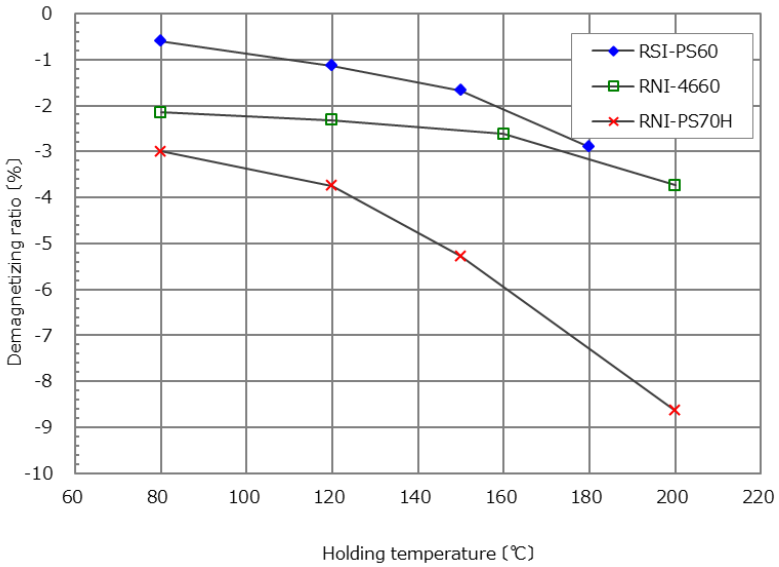
φ10×H7（Pc=2）



Variation against temperature of irreversible demagnetization ratio (flux)

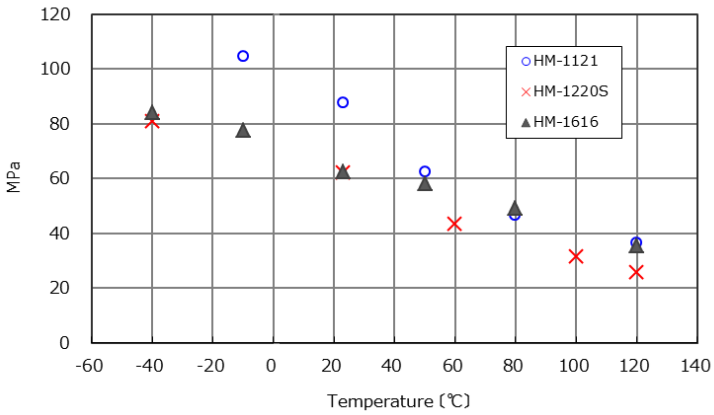
: after holding for 24 hours

φ10×H7（Pc=2）

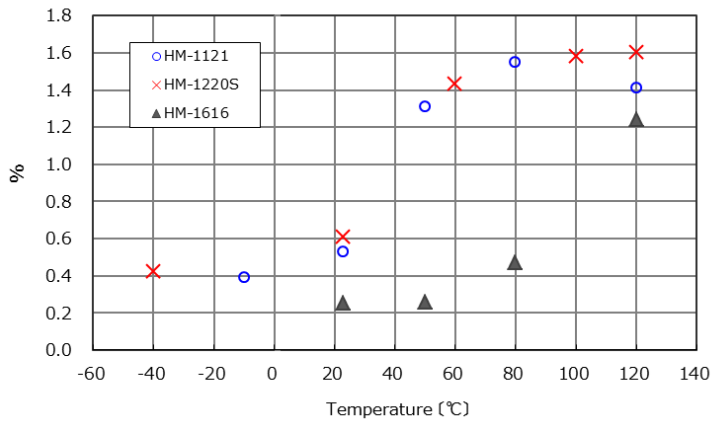


11. Thermal dependency

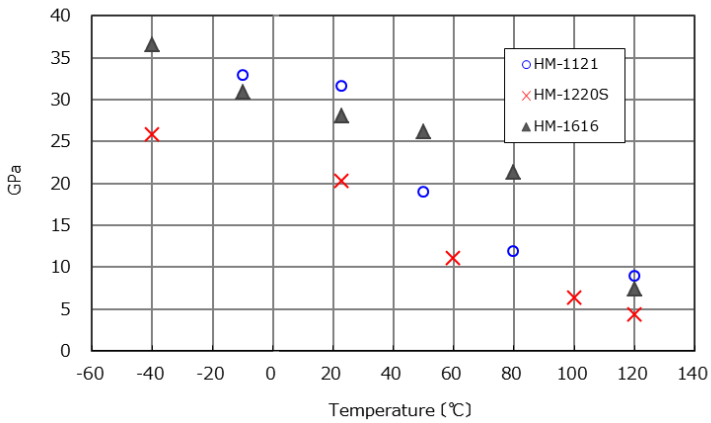
(1) Thermal dependency of tensile strength



(2) Thermal dependency of tensile elongation

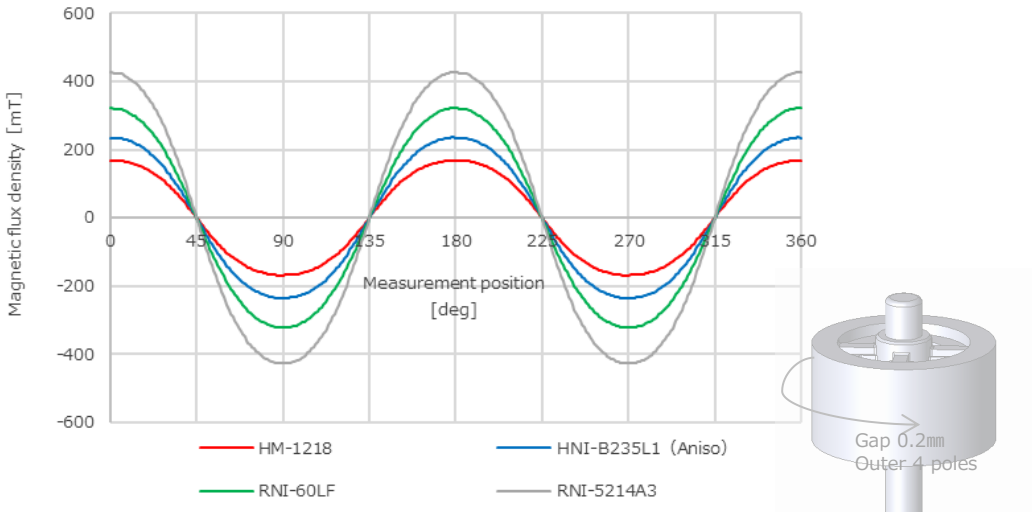


(3) Thermal dependency of tensile modulus

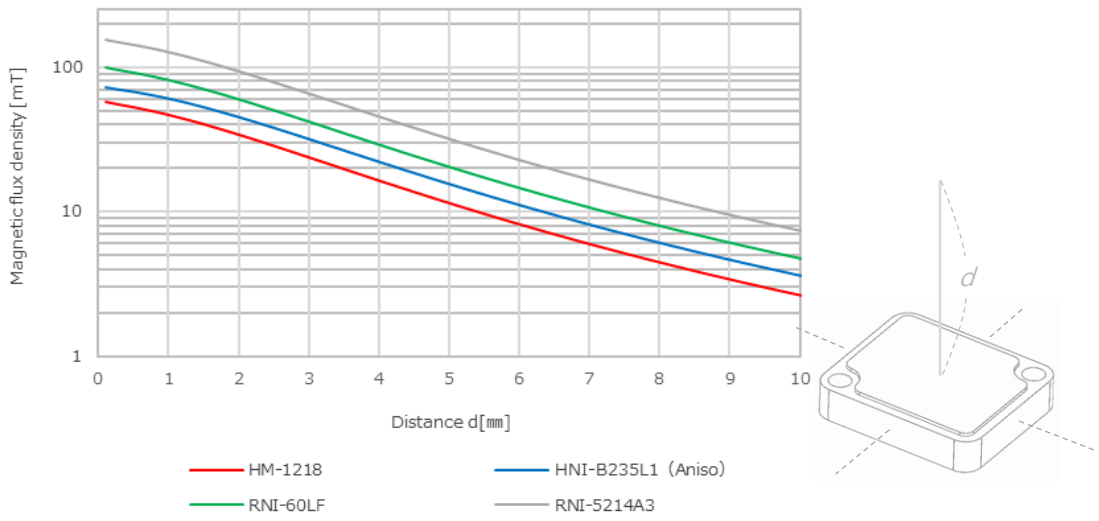


12. Molded product magnetic force (analytical value)

(1) Analyzed surface magnetic force values of molded products



(2) Distance from molded product surface and magnetic flux density analysis value



We will select the most suitable material according to the requirements of the molded product.

Soft magnetic compounds are the materials made with soft magnetic powders and thermoplastic resins suitable for injection molding.



1. Feature

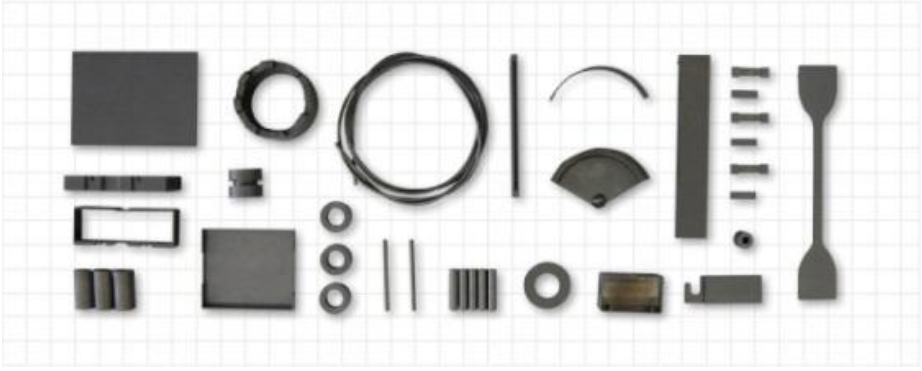
We control magnetic, electrical and physical properties, and moldability through our proprietary technologies in surface treatment, formulation and kneading.
We prepare various kinds of magnetic powders and resins, and can choose the best ones for each application.
Compounds can achieve compact, complex or thin shapes because of their excellent moldability.

Difference between hard and soft magnetic materials

Substance attracted to a magnet is called as magnetic substance. Among such substances, there are two types:
-Hard magnetic substances becomes a permanent magnet once attracted to magnet .
-Soft magnetic substances becomes a magnet when attracted to magnet but returns to the original state.

2. Applications

・Electromagnetic wave interference
Radiation noise suppression from CPU, PCB and cable of electric and electronic equipment
・Electromagnetic wave absorption
Radiation noise measure in chassis or connector of electric or electronic equipment
・Magnetic shielding
For RF-ID (FeliCa mobile phone, card), magnetic shielding of Induction Heating
・Magnetic core
Replacement of ferrite dust core for antenna, inductor and transformer
・Others
Sealant, sensor



Examples of molded and processed parts

3. Properties (values are typical and not guaranteed)

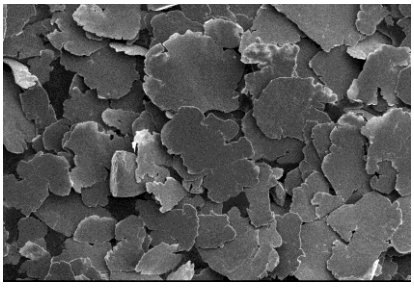
Soft magnetic compounds for injection molding

Filler Shape	Filler	Grade	Electromagnetic Properties			Physical Properties			Resin
			Magnetic Permeability μ' at 1MHz		Bs	Molded Density (ASTM D-792)	MFR (Melt Flow Rate)		
			φ13-8mm t5mm	φ20-10mm t0.6mm					
			-	-	T	g/cm^3	g/10min	measurement condition	
Sphere	FeSi	TM-6S70	16	-	1.20	5.6	800	330℃x60kg	PPS
Flake	FeSiAl	TS-1S50	30	60	0.45	3.9	130	280℃x60kg	PA6
		TS-2S50C	44	94	0.45	3.9	800	280℃x60kg	PA12
		TS-3S50	39	74	0.45	3.9	350	330℃x60kg	Heat Resistant PA
		TS-6S50	30	60	0.45	3.9	110	330℃x60kg	PPS
	FeSiCr	TS-2J50C	24	43	0.5	3.9	280	280℃x60kg	PA12

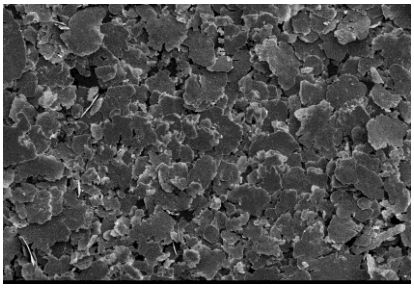
Measuring method
・Magnetic permeability : Impedance analyzer Agilent E4991A+16454A
・Bs : Automatic magnetization measuring equipment
・MFR : Flow tester （ orifice Φ1-L2mm, Preheating time 1min ）

Metal flake powder

Filler	Grade	Particle Size			Apparent Density	Tapping Density	Permeability (Coating Sheet Reference)	
		D10 μm	D50 μm	D90 μm	g/cm3	g/cm3	u' 3MHz	u" 13.56MHz
FeSiAl	SP-3B	10～25	35～55	≦150	0.30～0.50	0.80～1.20	160	60
FeSiCr	LP-5	5～10	15～30	≦80	0.30～0.45	0.55～0.90	60	1.3



SP-3B



LP-5

Soft magnetic compounds are regarded as “Dual-Use-items” and therefore certain export restriction do exist. Consequently, information about application and end-user is required prior to export from Japan.

High density compound DW series is a material for injection molding obtained by compounding tungsten and thermoplastic resin.
It is possible to injection mold parts that require a moment of inertia, high density and complex shape.

There are several grades that can be used to replace lead material, plastic parts with metallic textures, and for shielding radiation and X-ray.

1. Feature

Environmental friendly material that does not use environmental load substances such as lead or cadmium.
Can make parts smaller due to high density.

* Assuming density of iron is 7, volume of part becomes 54% if material with density of 13 is used.

Excellent texture, suitable for good appearance of parts that require good design.



2. Applications

(1) Applications requiring density (Vibration motor, fishing gear, balancer, weight, model gun)
(2) Applications requiring inertia (flywheel, G sensor)
(3) Applications requiring feel or texture (chassis, logo mark, plumbing products)











Mobile phone stand

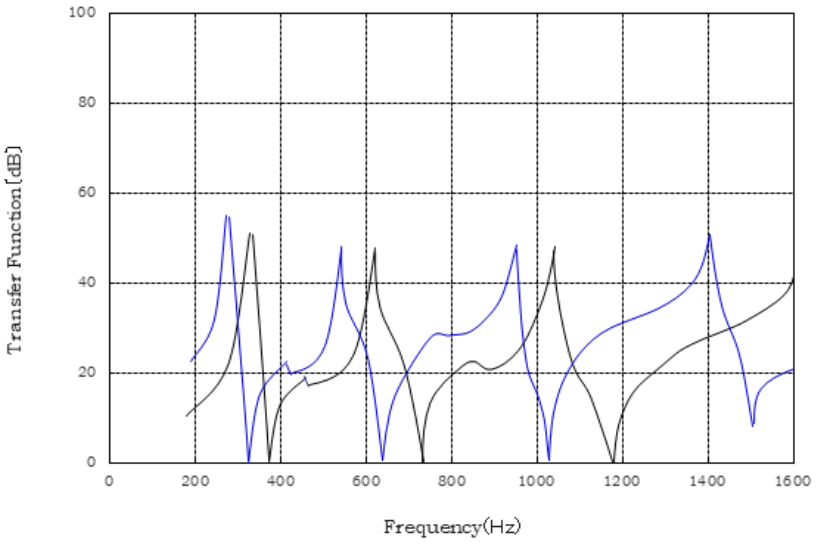


Acoustic isolation sheet

3. Properties (values are typical and not guaranteed)

Grade	Item / Unit	Molded Density g/cm^3	Flexural Strength MPa	Flexural Modulus GPa	IZOD Impact Strength kJ/m^2	Filler／Binder
DW-2070		7.0	68	4.6	25.5	W, ZnO／PA 12
DW-1100		10.0	139	11.3	42.2	W／PA 6
DW-1120		12.0	142	17.9	23.0	W／PA 6
DW-1130A		13.0	130	22.5	16.0	W／PA 6
Test Method (Ref. standard)		ASTM D-792	ASTM D-790		ASTM D-256 (Unnotched)	-

4. Vibration property (reference data)



By attaching high density material to a molded part, it is possible to shift the frequency characteristic lower. It is possible to adjust the frequency characteristic avoiding a specific frequency that will cause problems.

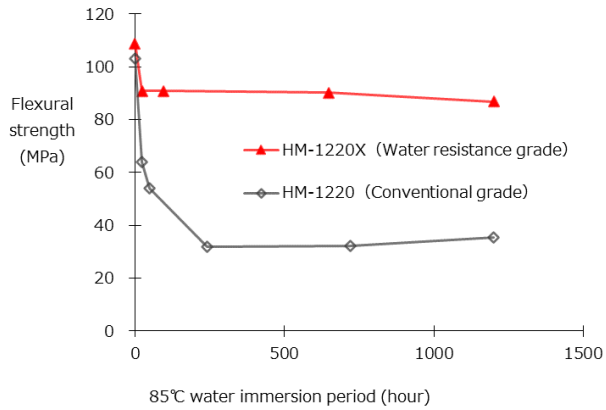
Heavy compounds are regarded as “Dual-Use-items” and therefore certain export restriction do exist. Consequently, information about application and end-user is required prior to export from Japan.

Only representative grades are shown in this catalog.
There are many grades not listed in this catalog that will meet your requirements.
Please let us know the characteristics and applications you require.
We will introduce the most suitable grade to your specific needs.

Example

Grade	HM-1220 (Conventional grade)	HM-1220X (Water Resistance grade)		
Item	Anisotropic	Anisotropic	Unit	Testing Method
Br	285	293	mT	JIS C2501**
bHc	199	200	kA/m	JIS C2501**
iHc	227	230	kA/m	JIS C2501**
BHmax	16.0	16.8	kJ/m^3	JIS C2501**
Molded Density	3.66	3.72	g/cm^3	ASTM D-792**
Melt Flow Rate MFR	110	100	g/10min	ASTM D-1238**
Flexural Strength	105	120	MPa	ASTM D-790**
Impact Strength IZOD	15.1	14.5	kJ/m^2	ASTM D-256** (Unnotched)
measurement condition of MFR	270℃×10kg	270℃×10kg	-	
Filler	Sr Ferrite	Sr Ferrite	-	
Binder	PA12	PA12	-	

Hot water resistance



Data not listed in the catalog, such as water/heat resistance and tensile strength, are available upon request.

*Please feel free to contact us through our website.
Visit us via QR code or URL.

Mate magnet

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